Ximera Euclid's Algorithm, GCD (section 36) COMP 61 (Discrete Math)

Question 1 *The goal of the Euclidean Algorithm is:*

Multiple Choice:

(a) *Computing* $a \bmod b$

(b) *Computing* $\gcd(a, b)$

(c) *Testing if* a *and* b *are prime*

(d) *Solving* $ax + by = 1$

(e) *Solving* $ax + by = \gcd(a, b)$

Question 2 *The goal of the Extended Euclidean Algorithm is:*

Multiple Choice:

(a) *Computing* $a \bmod b$

(b) *Computing* $\gcd(a, b)$

(c) *Testing if* a *and* b *are prime*

(d) *Solving* $ax + by = 1$

(e) *Solving* $ax + by = \gcd(a, b)$

Question 3 *If* a *and* b *are relatively prime, the smallest positive integer of the form* $ax + by$ *is:*

Select All Correct Answers:

(a) *0*

(b) *1*

(c) *2*

(d) *The GCD of* a *and* b

Question 4 *Which of the following are common divisors of 40 and 56?*

Select All Correct Answers:

(a) -4

(b) *1*

(c) *7*

(d) *8*

Question 5 *What is the gcd of 40 and 56?*

Multiple Choice:

(a) -4

(b) *1*

(c) *7*

(d) *8*

Question 6 *What is the gcd of 41 and 56?*

Multiple Choice:

(a) *1*

(b) *3*

(c) *7*

(d) *8*

Question 7 $\gcd(224, 140) = \boxed{?}$

Question 8 *The key sequence for computing* $\gcd(270, 72)$ *is:* $270, 72, \boxed{?}, \boxed{?}, \boxed{?}$

Question 9 *Find integers* x *and* y *such that* $ax+by = 1$, *where* $a = 88, b = 45$.

Multiple Choice:

(a) *No solution*

(b) $x = 5, y = -11$

(c) $x = 22, y = -43$

(d) $x = 37, y = -73$

Question 10 *There exists integers* x *and* y *such that* $ax + by = 1$ *for which of the following pair of values for* a and b

Multiple Choice:

(a) $a = 15, b = 105$

(b) $a = 3, b = 88$

(c) $a = 6, b = 78$

(d) $a = 10, b = 150$

Question 11 *There exists integers* x *and* y *such that* $ax + by = 1$ ______ a and b *are both prime numbers.*

Multiple Choice:

(a) *If*

(b) *Only if*

(c) *If and only if*

Question 12 *Let* a *and* b *be nonzero integers, let* $d = gcd(a, b)$. *If* e *is a common divisor of* a and b, *then...*

Multiple Choice:

(a) $e|d$

(b) $d|e$

(c) $d = e$

(d) *None of the above*

Question 13 *What is the gcd of 10 and* -5*?*

Multiple Choice:

(a) *1*

(b) *5*

(c) -5

(d) *10*

(e) *-10*